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## Lecture 4: Viruses, Worms, and Other Malware Types

### PART A: Computer Viruses

#### 1. What is a Computer Virus?

A **computer virus** is a type of **malicious software** that **attaches itself to a legitimate host file or program** and spreads when the infected file is executed by the user.

##### Key Characteristics

- Requires a host file
- Requires user action to activate
- Can replicate and spread
- Can modify or destroy data

#### 2. Purpose of Creating Viruses

Viruses are created for various reasons, including:

1. **Curiosity and Experimentation**
2. **Ego and Fame** (to demonstrate skills)
3. **Financial Gain** (data theft, extortion)
4. **Sabotage** (destroy systems or data)
5. **Political or Ideological Motivation**

#### 3. Indications of Virus Attack

Common symptoms of virus infection include:

- Slow system performance



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- Frequent system crashes
- Unexpected system restarts
- Corrupted or missing files
- Unusual pop-up messages
- Increased disk or CPU usage
- Antivirus alerts or disabled security software

#### 4. Stages of Virus Lifecycle

A computer virus typically goes through **four stages**:

##### 4.1 Dormant Phase

- Virus remains inactive
- Waits for specific conditions (date, event)

##### 4.2 Propagation Phase

- Virus replicates and infects other files or systems

##### 4.3 Triggering Phase

- Virus is activated when a condition is met

##### 4.4 Execution Phase

- Virus payload is executed
- Causes damage or performs malicious activity



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## **5. Types of Computer Viruses**

### **5.1 File Infector Viruses**

- Attach to executable files (.exe, .com)

### **5.2 Boot Sector Viruses**

- Infect boot sector or Master Boot Record (MBR)
- Activate during system startup

### **5.3 Macro Viruses**

- Written in macro languages (e.g., MS Word macros)

### **5.4 Polymorphic Viruses**

- Change code structure to evade detection

### **5.5 Metamorphic Viruses**

- Rewrite their entire code on each infection

### **5.6 Resident Viruses**

- Stay in memory and infect files as they are accessed

### **5.7 Non-Resident Viruses**

- Execute, infect selected files, then terminate



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## PART B: Computer Worms

### 6. How is a Worm Different from a Virus?

| Feature            | Virus    | Worm         |
|--------------------|----------|--------------|
| Host file required | Yes      | No           |
| User interaction   | Required | Not required |
| Self-replication   | Yes      | Yes          |
| Network spreading  | Limited  | Extensive    |

## PART C: Other Malware Types

### 7. Rootkits

A **rootkit** is malware designed to **hide its presence** and provide **privileged (root-level) access** to attackers.

#### Functions

- Conceal files and processes
- Disable security software
- Maintain persistent access

### 8. Potentially Unwanted Applications (PUAs)

PUAs are software that:

- Are not explicitly malicious
- Perform unwanted actions
- Often installed unknowingly



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## Examples

- Browser toolbars
- Ad injectors
- System optimizers with false alerts

## 9. Spyware

Spyware secretly **monitors user activities** and collects:

- Browsing history
- Login credentials
- Personal data

## 10. Keylogger

A **keylogger** records every keystroke typed on a system.

### Uses by attackers

- Capture passwords
- Steal credit card details
- Monitor communications

## 11. Botnets

A **botnet** is a network of **compromised computers (bots or zombies)** controlled by an attacker (botmaster).

### Uses

- DDoS attacks
- Spam campaigns
- Credential stuffing
- Cryptomining



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## 12. Fileless Malware

Fileless malware:

- Does not write files to disk
- Operates in memory
- Uses legitimate tools (PowerShell, Windows Management Instrumentation (WMI))

### Advantages for attackers

- Harder to detect
- Bypasses traditional antivirus

## 13. Summary Table

| Malware Type     | Key Feature                |
|------------------|----------------------------|
| Virus            | Requires host file         |
| Worm             | Spreads automatically      |
| Rootkit          | Hides presence             |
| PUA              | Unwanted but not malicious |
| Spyware          | Data spying                |
| Keylogger        | Keystroke capture          |
| Botnet           | Remote-controlled network  |
| Fileless Malware | Memory-based attack        |